

TITLECHAIN FOUNDATION

Technical Exhibits A-J - File No. S7-2026-30

Exhibit A - End-to-End 312 Spring Commons Reference Transaction

Reference: PPT-EZ-CA-0001 / 312 Spring Commons

Repository baseline: M5-AI-Sovereign-Commons 3a7b4b79b23399e9c362a124ca87f433a36321a3

Status: Submission exhibit; reference transaction remains demonstration-only.

A1. Transaction thesis

The simulation tests whether a government/public asset can move through a reconstructable chain without allowing a blockchain, token, wallet, AI model, or internal M5 label to replace the authoritative legal act.

PUBLIC ASSET IDENTIFIED

- > authoritative source/title evidence
- > jurisdiction + public owner
- > disposition pathway
- > accountable agency/office
- > authorized human(s)
- > required approvals
- > HOLD until external evidence
- > conveyance instrument
- > execution + recording
- > authoritative title state
- > project/stewardship rights
- > capital/debt/instrument layer
- > regulated recordkeeping where applicable
- > on/off-chain settlement + reconciliation
- > operating contracts/productive assets
- > continuing compliance/evidence
- > correction / transfer / redemption / exit

Across every stage:

HUMAN/ENTITY -> CAPACITY -> AUTHORITY -> POLICY -> APPROVAL -> EXECUTION -> AUTHORITATIVE
RECORD -> RECEIPT

A2. Separate record domains

Domain	What it establishes	Must not be silently replaced by
Physical/public title	Recorded ownership and conveyance state	token balance, wallet, simulation
Government authority	Power to dispose/approve/sign	AI output, namespace, credential alone
Stewardship/operations	Bounded operating rights	fee title
Debt/capital	Financing obligations and rights	property title
Securities, if any	Issuer/holder rights and restrictions	project membership
Transfer-agent record	Registered holder state where applicable	wallet balance alone
Banking/settlement	Authoritative movement of funds on the applicable rail	blockchain evidence alone
Blockchain/evidence	integrity, event, provenance, reconciliation evidence	legal authority
Productive assets	equipment/services/output state	title to the underlying land

A3. Demonstrated control behavior

The repository's fit-gap review correctly treats SIMULATED_PASS as an internal rule result and PENDING_EXTERNAL as missing authoritative evidence. A simulation cannot self-advance because code passed.

The reference transaction therefore uses the state path:

IDENTIFIED -> SOURCE_VERIFIED -> AUTHORITY_MAPPED -> APPROVALS_REQUIRED -> HOLD -> APPROVED
-> EXECUTED -> RECORDED -> AUDITED

Exception states: REJECTED, REVOKED, DISPUTED, SUPERSEDED.

A4. Financing and settlement

The simulation keeps capital provenance and settlement separate from title. Material funding events should preserve entity, authority, source-of-funds, jurisdiction path, regulated route, destination, governing instrument, and reconciliation evidence.

Illustrative architecture:

capital source -> lawful vehicle/account -> eligibility/authority -> subscription/financing instrument -> bank or permitted digital rail -> escrow/project account -> draw approval -> payee -> settlement evidence -> reconciliation -> project/evidence update

A successful blockchain event does not prove that bank settlement occurred. A bank transfer does not prove that a title transfer occurred. Each authoritative domain remains independently evidenced and linked.

A5. Mandatory disclosure

This reference transaction demonstrates the technical, authority, evidence and approval workflow. It does not represent government authorization, agency participation, completed title transfer, live settlement, regulated-provider appointment, or independent validation unless corresponding evidence is explicitly identified.

Exhibit B - Repository and Contract Architecture Crosswalk

Pinned baselines:

- M5-AI-Sovereign-Commons: 3a7b4b79b23399e9c362a124ca87f433a36321a3
- icsn-standards: a0a7341213dbf4abad5fa0aeb46b43e7b95c6660

Layer / artifact	Purpose in SEC demonstration	Authority boundary
pilots/312-spring-commons/README.md	Full public-asset reference transaction, capital provenance, title/right separation	Draft; external legal records control
pilots/312-spring-commons/INITIATIVE-FIT-GAP-REVIEW.md	Evidence gates; simulated pass vs pending external	Code success is not approval
docs/PILOT-SEC-RFI-CROSSWALK.md	Maps public pilot to SEC recordkeeping questions	Research crosswalk, not SEC position
docs/M5CANON-FUNCTION-NAMESPACE.md	Deterministic function/authority namespace	M5Canon enforces supplied authority; does not create it
docs/M5AGT-AUTHORITY-AND-ACTIVATION.md	Agent activation/delegation constraints	Agent cannot exceed principal
docs/M5-CREDENTIAL-TRUST-AND-JURISDICTION-BINDING.md	Entity/role/credential/jurisdiction evidence stack	Credentials are evidence, not licenses
docs/M5-CONFORMANCE-AND-NAMESPACE-BOUNDARIES.md	Namespace and conformance boundaries	Friendly namespace is not governmental authority
docs/PEOPLES-TRUST-PUBLIC-PILOT.md	Repeatable farmland conformance simulation	Synthetic/proposed until external evidence
shadow-m5index/README.md	Public asset/debt/ownership research layer	Research state separate from canonical authority
shadow-m5index/reports/SHADOW-CAMEL-REPORT-Q3-2026.md	Published public-data systemic-risk research	Not official CAMELS/regulator finding
docs/SHADOW-M5INDEX-PUBLIC-INGESTION-AND-QR.md	Public contribution, evidence, QR/asset indexing	Submission cannot self-promote to canonical
ICSN docs/M5MST-ASSET-MINTING-AND-TITLECHAIN-REGISTRATION.md	Origination/title-registration event model	Mint/origination authority != transfer authority
ICSN docs/TRANSFER-AGENT-DIGITAL-ASSET-REFERENCE-ARCHITECTURE.md	Known entity -> human -> bounded agent -> recordkeeping	Regulated TA remains accountable
ICSN docs/SWIFTBRIDGE-INTEROPERABILITY.md	Maps legacy identifiers/rails without discarding originals	Not SWIFT/BIS endorsement or replacement
ICSN docs/M5-GLOBAL-INDEX-AND-EXCHANGE.md	Provenance-aware public economic intelligence	Not a securities exchange/ATS/trading venue
ICSN architecture/M5-L0-L8-ZK-REFERENCE-ARCHITECTURE.md	Layered proof/credential/reference architecture	Proof system does not create authority
ICSN Article 12 CER mapping	Keeps CER control distinct from title/authority	Control != title != authority

Review rule

Repository presence proves that an artifact exists at the pinned commit. It does not prove an external fact, agency approval, production deployment, legal conclusion, or independent replication.

Exhibit C - M5 Classification to Regulatory Authority Routing

Purpose

M5 classification is a machine-readable routing aid. It does not make a governmental jurisdictional determination.

ASSET / RIGHT

-> M5 CLASSIFICATION

-> TRANSACTION / EVENT

-> ACTOR + CAPACITY

-> JURISDICTION

-> EXTERNAL LEGAL CHARACTERISTICS

-> POTENTIALLY APPLICABLE AUTHORITIES

-> REGULATED FUNCTION

-> REQUIRED LICENSE / APPROVAL / RECORD / EVIDENCE

-> AUTHORIZED HUMAN / BOUNDED AGENT

-> HOLD / ALLOW / REJECT

-> EXECUTION

-> AUTHORITATIVE RECORD

-> RECEIPT

Routing matrix

Event / characteristic	Potential pathway to evaluate	Authoritative determination remains with	Fail-closed evidence
Security issuance/transfer/holder record	Federal/state securities + transfer-agent requirements	SEC, applicable state law, courts, authorized regulated entities	instrument analysis, issuer authority, TA status where required, restrictions
Commodity/derivatives activity	Commodity/derivatives requirements	CFTC and applicable law/market authorities	product/activity analysis, registration/venue status where required
Bank/payment/settlement event	Banking/payment/settlement requirements	applicable federal/state banking/payment authorities	account owner, authorized signer, rail, settlement record
AML/BSA-relevant regulated activity	AML/BSA controls	FinCEN plus applicable functional regulator(s)	covered entity/activity, policy, screening/KYC evidence as required
Real-property conveyance	State/local property/title/recording	applicable state/local law, recorder/clerk/court	deed/instrument, authorized signers, recording evidence
Government asset disposition	Public-property/disposition rules	owning agency and applicable statutory	disposition authority, approvals, restrictions,

		authorities	executed instrument
Tax event	Federal/state/local tax	IRS and applicable state/local tax authorities	transaction facts, classification, filing/payment evidence
Professional act	Licensing/professional rules	applicable licensing authority	current license/appointment/scope
Ambiguous or mixed event	Multi-agency/legal review	relevant authorities/counsel	REGULATORY_HOLD until resolved

Required machine states

- ROUTE_IDENTIFIED
- ROUTE_MULTI_AUTHORITY
- EXTERNAL_CLASSIFICATION_REQUIRED
- REGULATORY_HOLD
- LEGAL_REVIEW_REQUIRED
- AUTHORITY_EVIDENCE_VERIFIED
- ROUTE_SUPERSEDED

No internal classification may self-promote to GOVERNMENT_APPROVED.

Exhibit D - Government Authority and Approval Matrix

Question	Required record
Who owns/administers the public asset?	authoritative public/title record
Which office has disposition authority?	statute/regulation/delegation/agency record
Which human may act?	current appointment/role/signature authority
In what capacity?	office/entity role and scope
Which approvals are prerequisites?	approval matrix with responsible authority
Can an agent assist?	explicit bounded delegation
What is the controlling instrument?	deed, resolution, contract, approval, permit, etc.
What happens if evidence is missing?	HOLD/REJECT
What changed after approval?	versioned authority/evidence state
What record proves execution?	authoritative system-of-record update + receipt

Approval receipt minimum

who | acting_for_whom | capacity | jurisdiction | authority_source | scope | decision | timestamp | credential_state | evidence | resulting_state | revocation/supersession

Exhibit E - Capital, Debt, Securities and Settlement Architecture

Separation principle

A project may contain multiple economic and legal layers. They must not be collapsed into one token or one ledger entry.

PHYSICAL / PUBLIC TITLE

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STEWARDSHIP / OPERATING RIGHTS

|

PROJECT ENTITY / CONTRACTUAL RIGHTS

|

DEBT / EQUITY / OTHER CAPITAL INSTRUMENTS

|

SECURITY INSTRUMENT + TRANSFER AGENT (WHEN APPLICABLE)

|

BANK / ESCROW / PAYMENT / PERMITTED DIGITAL RAILS

|

RECONCILIATION + EVIDENCE

Required fields for a material capital event

- source entity and accountable human;
- authority and account purpose;
- instrument/legal terms;
- asset/economic classification;
- external legal/regulatory classification where required;
- jurisdiction path;
- source-of-funds evidence appropriate to the regulated context;
- destination account and owner;
- rail/provider;
- approval thresholds;
- settlement confirmation;
- reconciliation result;
- exceptions/corrections;
- evidence receipt.

On/off-chain rule

Onchain evidence may attest to an event, instruction, state reference, or reconciliation. It does not silently replace the authoritative bank, escrow, transfer-agent, recorder, issuer, or other legally controlling record.

Exhibit F - Public Evidence and Edge Participation Network

Objective

Activate local and professional participants as evidence contributors without turning contribution into authority.

Participant	Useful contribution	Authority boundary
County recorder/clerk	recorded instruments, indexes, official copies	official records control according to applicable law
Title company	title search, exceptions, chain/encumbrance work	report/insurance scope controls; not government title creation
Notary	notarial act evidence	only within commissioned authority and applicable law
Assessor	parcel/assessment records	assessment record is not necessarily title
Government disposition office	inventory, status, disposition evidence	agency authority and official records control
Transfer agent	securityholder/transfer records	registered TA responsibility where applicable
Bank/escrow/payment provider	account/settlement evidence	provider's authoritative records control
Licensed professional	scoped attestation	credential, jurisdiction and engagement scope control
Researcher/citizen	public-source discovery and challenge	submission is not canonical authority

Evidence path

DISCOVERED -> SUBMITTED -> SOURCE_VERIFIED -> AUTHORITY_VERIFIED -> REVIEWED -> CANONICAL_WHERE_APPLICABLE

Exception states: DISPUTED, CORRECTED, SUPERSEDED, REJECTED.

Every contribution should preserve contributor/capacity, source, jurisdiction, document type, evidence class, timestamp, version, signature/proof if applicable, linked asset/event, review status, and dispute/supersession state.

Exhibit G - SHADOW M5Index and SHADOW CAMEL Public-Data Pipeline

Existing public artifact

The repository contains SHADOW CAMEL Report - Q3 2026, described as a first-edition public-research release using public sources, methodology v0.1, confidence labels, charts, a 36-record asset appendix, and public contribution guidance.

The report's preliminary composite reading is 1.8 / 5 - Band 2 (Stable / Watch), confidence B. That is the Foundation report's own public-research output. It is not reproduced here as an official or independent assessment.

Mandatory boundary

SHADOW CAMEL is not:

- an official CAMELS rating;
- confidential supervisory information;
- a bank examination;
- a regulator finding;
- an investment recommendation;
- a credit rating;
- a legal conclusion.

Public-data pipeline

government/public asset sources

+ county/registry evidence

+ CRE/farmland/public-property datasets

+ debt/maturity/ownership evidence

+ public banking/market data where lawful

-> provenance capture

-> evidence classification

-> source verification

-> asset/right/debt graph

-> SHADOW M5Index research

-> SHADOW CAMEL public-data methodology

-> public challenge/correction

-> versioned superseding edition

Edge participation

The repository already provides issue pathways to nominate a public asset, add registry/public-record evidence, and challenge/correct an index record. A submission enters review; it does not become canonical merely because it was contributed.

Only approved public-safe aggregate events should flow to M5 Global economic-intelligence outputs.

Exhibit H - Industry Architecture Crosswalk

Source	Documented proposition	Convergence	Remaining architecture question
Plume Network (Sept. 21, 2026)	tokenized vault roles, programmable compliance, TA modernization, DLT wallet-address treatment	electronic records and automated compliance	wallet/control vs authority; attestation provenance; migration; freeze/seize authority
Ceres Coin TA (Sept. 22, 2026)	authoritative regulated register for holder-rights parity	authoritative record primacy	reconciliation with token state; successor continuity
NeuFin (Sept. 22, 2026)	material-change revalidation; records linking prior state, authority boundaries and outcomes	lifecycle/revalidation	deterministic validity windows and fail-closed action
Gene Deyev (Sept. 10, 2026)	source-linked, timestamped, evidence-typed issuance record; version preservation	provenance/version history	authority lifecycle and machine-action receipts
TitleChain Foundation	control/title/authority/ notice/evidence separation; fail-closed authority	implementation-neutral evidence architecture	external review and independent replication

Boundary: Industry proposal != SEC position != Foundation analysis != machine test != independent validation != adopted standard.

Exhibit I - Failure, Recovery and Successor-State Tests

Test	Expected behavior
Valid key + revoked authority	HOLD/REJECT; no inference from cryptographic validity
Missing agency approval	HOLD; simulation cannot create approval
Expired credential	HOLD and revalidation
Jurisdiction mismatch	HOLD/REJECT and authoritative resolution
False/friendly namespace	no governmental authority inferred
Wallet/TA record conflict	HOLD; reconcile to authoritative record
Transfer journal/MSF inconsistency	record difference; investigate/reconcile
Stale/disputed attestation	no authority inference; review
Unauthorized freeze/seize	REJECT; preserve attempted action evidence
Provider/chain failure	successor-state migration without changing rights/authority
Key compromise	revoke/replace key; preserve authoritative history
Cryptographic deprecation	migration receipt; no silent state change
TA succession	portable records, authority, reconciliation, exceptions
AI agent exceeds delegation	DENY; preserve evidence receipt
Bank rail says paid, chain says pending	no title/holder change until authoritative conditions reconcile

Cryptographic resilience invariant

Technical control does not manufacture authority, and the authoritative record must survive the technology used to represent it.

Exhibit J - Demonstrated / Built / Pipeline / External / Not Yet Demonstrated

Demonstrated in Foundation simulation

- fail-closed authority/evidence concepts;
- separation of title, stewardship, capital, regulated recordkeeping and settlement domains;
- Spring Commons structured transaction and diligence architecture;
- People's Trust repeatable conformance architecture;
- SHADOW public research/reporting and contribution pathways;
- machine-readable schemas/tests/reference artifacts identified in the pinned repositories.

Built in public repository

- M5Canon authority controls;
- M5AGT bounded activation/delegation materials;
- credential/jurisdiction binding;
- M5MST origination/title-registration reference;
- transfer-agent digital-asset reference architecture;
- Article 12 CER mapping;
- SwiftBRIDGE interoperability;
- M5 Global Index and Exchange research architecture;
- SHADOW M5Index;
- SHADOW CAMEL Q3 2026 report;
- Spring Commons and farmland simulations.

Pipeline / proposed public review

- expanded SEC supplemental exhibits;
- regulatory-routing schema and receipts;
- cryptographic dependency capture/failure profile where not already merged;
- broader edge-participant ingestion profiles;
- additional independent conformance implementations.

Requires external participation/evidence

- government disposition authorization;
- recorder/title confirmation;
- regulated transfer-agent appointment;
- bank/escrow/live settlement;
- counsel/legal conclusions;
- agency/regulator determinations;
- licensed-professional attestations;
- independent replication.

Not yet demonstrated

- completed government-to-public title transfer;
- live securities issuance/transfer under the Spring reference transaction;

- live project debt closing;
- live on/off-chain settlement reconciliation for the reference transaction;
- government or SEC endorsement;
- independent validation of the complete end-to-end transaction.

Mandatory disclosure: A repository artifact, test pass, internal label, blockchain transaction, AI output, public contribution, or Foundation publication does not itself establish external authority or legal effect.